8.4.5 The Efficiency of a Thermostat and "Gentle" Thermostats

To understand the reason for this fundamental, qualitative difference between Nosé-Hoover-Langevin and Langevin dynamics, and to compare it with Velocity Rescaling, one may study a quantity that relates the rate of convergence to equilibrium to the rate of growth of the error in the autocorrelation function. In [225], this precise quantity is introduced and termed the *efficiency* of the thermostat:

$$\text{efficiency} = \frac{\text{rate of convergence to equilibrium}}{\text{rate of growth of perturbation of dynamics}}.$$

The rate of convergence to equilibrium is quite difficult to calculate in general, as it would typically relate to the spectral gap of the diffusion. In light of the observations of Fig. 8.3, one may choose to estimate this in terms of the rate of convergence of the average kinetic energy to the target temperature.

The rate of accumulation of error due to the thermostat is also difficult to quantify analytically, since it requires calculation of the error in an autocorrelation function over a time interval which may not be short compared to the simulation stepsize, particularly for slow-relaxing systems. Again, an estimate is obtainable by considering the rate of error accumulation at short times (obtained by Taylor expansion of the exact and approximate autocorrelation functions). The result of these two estimates is then an analytical estimate for the efficiency that relates various schemes for various values of the parameters involved.

An interesting observation is that, while for both Langevin dynamics and Velocity Rescaling the error accumulation is linear in time, the same is not true for Nosé-Hoover-Langevin: the error in NHL increases initially in proportion to t^2; the instantaneous diffusion rate is inherited from the system to which it is applied. This means that, while NHL cannot be used to model imposed target diffusion rates, its influence on dynamics within the setting of molecular dynamics thermostatting is also very slight: it is a *gentle* thermostat. The Velocity Rescaling method is also gentle in another sense: the rate of growth of error is small. Estimating the efficiency for both Langevin and Velocity Rescaling is similar—we find the ratio of the estimated rate of convergence of kinetic energy to the estimated rate of linear error growth in the velocity autocorrelation function. For Nosé-Hoover-Langevin, which has quadratic-in-time error growth, we determine the characteristic time until convergence (the reciprocal of the exponential rate of convergence) and from this determine how much perturbation accumulates in the velocity autocorrelation function.

In Table 8.1 we show the comparison of the relevant estimates of the above quantities, including the efficiency, for different choices of thermostat.

Table 8.1 Estimated rates of convergence, perturbation growth, and efficiency for different methods

Thermostat	Convergence rate	Error (to lowest order in t)	Efficiency
Langevin	γ	γt	≈ 1
Velocity Rescaling	$4\,\frac{\mathrm{Av}_\beta\{K^2\Phi(K)\}}{\mathrm{TN_d k_B}}$	$\frac{2}{\mathrm{N_d}}\mathrm{Av}_\beta\{K\Phi(K)\}\,t$	$O(\mathrm{N_d})$
VR with $\Phi(K)=\frac{\gamma_{\mathrm{VR}}}{2K}$	γ_{VR}	$\frac{\gamma_{\mathrm{VR}}}{\mathrm{N_d}}t$	$\approx \mathrm{N_d}$
NHL	$\sqrt{\frac{\mathrm{N_d}k_\mathrm{B}\mathrm{T}}{\mu}}$ (at critical damping)	$\frac{k_\mathrm{B}\mathrm{T}}{2\mu}t^2$	$\approx 2\mathrm{N_d}$

8.5 Non-Newtonian Dynamics-Based Iterations for Molecular Sampling

Given that we can decompose the stochastic dynamics into deterministic part and stochastic process, and approximately or exactly solve the parts, it is common to iterate the maps for each building block sequentially in such a way that the calculation approximates the dynamics of the system. This is, very likely, an unnecessary restriction on the sampling strategy. The crucial things that effect the ultimate success of the method are that the parts themselves preserve the measure (or approximately so), the stages are volume preserving, and that the stochastic processes are sufficiently strongly coupled so that the overall iteration is ergodic.

There is essentially no need to use maps based on Newtonian deterministic dynamics. There are many alternative dynamics models which leave invariant the canonical measure, going well beyond those considered so far. To illustrate this with one intriguing example, we explore here the stochastic dynamical system in $2\mathrm{N_c}+1$ variables defined by

$$\mathrm{d}\boldsymbol{q} = \xi M^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.33}$$

$$\mathrm{d}\boldsymbol{p} = -\gamma\boldsymbol{p}\mathrm{d}t + \sigma\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}, \tag{8.34}$$

$$\mathrm{d}\xi = -[(\boldsymbol{M}^{-1}\boldsymbol{p})\cdot\nabla U(\boldsymbol{q})/\mu + \gamma'\xi]\mathrm{d}t + \sigma'\mathrm{d}Z, \tag{8.35}$$

where $\mathbf{W}$ is a vector of $\mathrm{N_c}$ Wiener processes, Z is a scalar Wiener process and $\sigma = \sqrt{2\mathrm{k_B T}\gamma}$, and $\sigma' = \sqrt{2\mathrm{k_B T}\gamma'/\mu}$ this system preserves the augmented distribution with density

$$\tilde{\rho}(\boldsymbol{q},\boldsymbol{p},\xi) = e^{-\beta(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2+U(\boldsymbol{q})+\mu\xi^2/2)}.$$

For this system, it is possible to show that the Hörmander condition holds. Setting all parameters to 1 and taking $\boldsymbol{M}=\boldsymbol{I}$ (without loss of generality in this demonstration) we write our SDE in the form $\mathrm{d}\varphi = D + \sum_{i=1}^{\mathrm{N_c}+1} Y_i \mathrm{dW}_i$, where φ is the vector of all phase variables, $Y_i = \boldsymbol{e}_{\mathrm{N_c}+i}$, for $i = 1,\ldots,\mathrm{N_c}+1$, is the $i+\mathrm{N_c}$th basis function in